

WILLOW RUN, MI, USA

WMO: 725376

Lat: 42.233N

Lon: 83.533W

Elev: 237

StdP: 98.51

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14853

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.5	-14.6	-22.2	0.5	-17.0	-19.0	0.7	-13.5	12.5	-4.9	11.4	-2.2	3.7	220	0.556

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	33.3	23.3	31.8	22.6	30.1	21.6	25.1	30.9	24.0	29.3	23.1	28.0	5.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.1	18.4	28.3	22.4	17.6	27.2	21.4	16.5	26.1	77.2	30.6	73.1	29.4	69.5	28.2	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	9.5	8.5	DB	-21.2	35.4	4.2	2.0	-24.2	36.8	-26.6	38.0	-29.0	39.1	-32.0	40.5	(4)
(5)			WB	-21.5	26.8	4.1	1.1	-24.5	27.5	-26.8	28.2	-29.1	28.8	-32.1	29.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	-3.9	-2.7	2.5	9.2	15.6	21.0	23.1	22.1	18.2	11.5	5.0	-0.6	(6)	
	DBStd	10.68	6.35	5.91	5.90	4.89	4.82	3.81	3.30	3.06	4.16	4.67	4.96	5.45	(7)	
	HDD10.0	1642	431	357	243	71	7	0	0	0	1	36	164	332	(8)	
	HDD18.3	3441	688	589	490	275	114	18	3	5	53	217	402	588	(9)	
	CDD10.0	1697	1	1	12	48	180	329	406	374	246	84	13	2	(10)	
	CDD18.3	455	0	0	1	2	29	97	151	121	48	6	0	0	(11)	
	CDH23.3	4370	0	0	8	38	336	938	1499	1052	442	57	1	0	(12)	
	CDH26.7	1532	0	0	2	6	108	346	581	340	139	10	0	0	(13)	
(14) Wind	WSAvg	4.0	4.7	4.6	4.6	4.7	3.8	3.4	3.2	3.1	3.2	3.9	4.4	4.5	(14)	
(15) Precipitation	PrecAvg	818	53	45	53	72	93	87	84	84	74	64	58	53	(15)	
	PrecMax	1053	111	103	106	142	186	162	167	163	147	172	139	112	(16)	
	PrecMin	663	14	16	10	11	24	30	38	11	22	1	17	16	(17)	
	PrecStd	104	26	21	28	30	46	32	32	38	36	33	28	24	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.1	16.1	23.7	27.4	32.4	34.9	35.9	34.1	33.4	27.9	20.7	16.3	(19)	
		MCWB	12.4	11.9	16.7	17.8	21.2	23.6	24.9	23.8	23.2	20.3	13.9	13.9	(20)	
	2%	DB	10.8	11.3	18.8	23.8	29.7	32.5	33.6	31.8	30.5	24.7	17.4	12.4	(21)	
		MCWB	8.9	8.3	12.7	15.3	20.1	22.5	23.6	23.2	21.4	18.0	13.0	10.2	(22)	
	5%	DB	7.4	7.7	15.1	21.1	27.3	30.6	31.8	30.1	27.8	22.2	15.4	9.4	(23)	
		MCWB	5.4	4.9	10.6	13.8	19.1	21.5	22.9	22.2	20.4	16.4	12.1	7.1	(24)	
	10%	DB	4.2	5.0	12.0	18.1	24.6	28.5	29.9	28.5	25.9	19.5	12.7	6.6	(25)	
		MCWB	2.6	2.8	8.2	12.0	17.7	20.6	21.8	21.4	19.2	15.0	9.6	4.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.3	12.7	17.1	19.3	22.9	25.1	26.6	26.0	24.9	21.7	16.1	13.9	(27)	
		MCDB	13.7	14.7	23.4	25.8	29.4	32.4	32.9	31.8	31.1	26.6	18.3	16.3	(28)	
	2%	WB	8.4	8.5	13.9	16.8	21.5	23.9	25.2	24.7	23.0	19.2	14.1	10.4	(29)	
		MCDB	9.5	11.2	17.3	21.4	27.7	30.3	31.3	29.5	27.6	22.8	16.5	12.2	(30)	
	5%	WB	4.9	5.0	11.4	14.8	20.2	22.9	24.1	23.6	21.7	17.5	12.1	7.3	(31)	
		MCDB	6.6	7.5	14.8	19.9	25.6	28.6	29.4	27.9	25.9	21.0	14.8	9.3	(32)	
	10%	WB	2.5	2.9	8.0	12.7	18.7	21.8	23.1	22.6	20.4	15.5	10.0	4.5	(33)	
		MCDB	4.0	4.8	11.4	17.7	23.6	26.8	27.9	26.8	24.3	19.0	12.7	6.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.7	8.6	10.1	11.8	12.1	11.8	11.8	11.3	12.1	10.7	8.8	7.0	(35)	
		MCDBR	9.7	11.6	15.1	16.3	14.8	14.1	13.2	12.6	14.5	14.0	12.2	9.7	(36)	
	5% WB	MCWBR	8.5	9.3	10.0	9.7	7.7	6.4	5.7	5.8	7.1	8.1	9.1	8.0	(37)	
		MCWBR	9.3	11.2	13.9	14.3	13.1	12.0	11.3	10.6	11.9	12.1	11.0	9.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.299	0.337	0.354	0.401	0.431	0.450	0.441	0.433	0.388	0.356	0.317	0.311	(40)	
		taud	2.393	2.264	2.337	2.227	2.194	2.179	2.227	2.239	2.344	2.417	2.489	2.392	(41)	
	Ebn at Noon		844	856	889	867	847	829	832	827	842	828	820	792	(42)	
		Edh at Noon	79	106	112	134	143	146	137	132	110	91	73	74	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.55	2.43	3.52	4.61	5.37	6.04	6.08	5.27	4.27	2.78	1.75	1.22	(44)	
	RadStd		0.14	0.31	0.35	0.40	0.47	0.39	0.31	0.20	0.37	0.24	0.23	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.88	N/A	N/A	+0.76	N/A	N/A	N/A	-228	+211	+127
(47)	Regional (63 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+83	N/A

Nomenclature: See separate page